TASK-8 FlyEco

Group 15

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1. Introduction:

Ambitious environmental targets, such as those set by the Air Transport Action Group within its Waypoint 2050 targets, are driving ambitions for a potentially transformative transition towards sustainable propulsion technologies in aviation. Potential alternatives include aero-engine electrification, which marries advanced electrical systems and hybrid-electric propulsion together with conventional propulsion. The idea is to preserve high performance and reliability while dealing with major challenges: reducing greenhouse gas emissions, improving fuel economy, and operating costs. Aero-engine electrification allows for new design options in aircraft that can be considered due to improved utilization of energy with less dependence on fossil fuels. Hybrid-electric propulsion systems combine gas turbines with electric motors, Solid Oxide Fuel Cells (SOFC), and thermal management systems into one integrated package to optimize power supply at different phases of flight. Electricity's introduction into aviation opens the aircraft design space, enabling such truly innovative concepts of aircraft and propulsion system integration as box-wing and blended-wing body aircraft, together with boundary-layer ingestion. That is, the purpose is to use more traditional aircraft designs to identify the medium-term possibilities and difficulties with electric propulsion systems.

The project covers an iterative design process in which continuous sizing of the aircraft, the propulsion system, and the components within can be done. In the frame of this project, two different integration concepts were proposed: Integrated SOFC with Turbofan Engine and Partially Nacelle-Integrated Propulsion. The likely problems and compromises between the two concepts are under analysis and discussion.

1.1 Why Aeroengine Electrification and Hybridization?

- 3% of the worlds CO_2 emissions comes from Aviation.
- Average CO2 emissions per passenger kilometers (pkm) is 70-100 g CO_2/pkm today flying by aircraft which is significantly higher than compared to a car.
- These emissions include Noise, Ozone, Soot, NOx, and mainly hydrocarbons (CO2) resulting in hazardous impacts to nature and humans. European Union (EU) aims to significantly reduce these potential emissions from the Aviation industry till year 2050.
- Aeroengine hybridization includes various propulsion systems such as electric propulsion systems, H2-based propulsion systems, and the Hybrid Electric Propulsion System.
- Definition of Hybridization factors defining the type of Aeroengine:

  $H_P=P_{EM} / (P_{EM}+P_{ICE})$ and $H_E=E_{Elect} / (E_{Elect}+ E_{Fuel})$
  H_E (Energy Hybridization) = Ratio of the installed electric energy to the total installed energy including chemical fuel.
  H_P – (Power Hybridization) = Ratio of the total installed electric motor shaft power to the total installed shaft power.
- When H_P= 0 (Conventional Gas Turbine Aeroengine) and H_P=1 (All Electric Aeroengine).
- Our aim involves developing such a hybrid electric Aeroengine integration concept with 0,0 < H_P < 1,0 and 0,0 < H_E < 1,0 to maximize efficiency, and performance with minimal weight addition. The proposed integration concepts couple the conventional gas turbine with fuel cell and electrical components, unlike the fully electric aeroengine which has its own limitations.

2. Problem Statement:

Combine cycle concepts that combine gas turbines, fuel cells, and electrical components used to enhance conventional gas turbine performance. Although the increase in efficiency and performance comes at the cost of additional mass and volume. To develop two or more engine integration concepts for a nacelle-integrated gas turbine-SOFC powertrain with power provisioning of 1 MW.

3. Literature Review:

3.1 Hybrid Electric Propulsion in Aviation:

Hybrid-electric propulsion systems mark the new generation in aviation by fusing conventional fuel-based engines with electric propulsion technologies. With higher fuel efficiency, lower emissions, and therefore reduced costs of operation, these systems mean just what modern aviation needs with the growing demand for sustainability in aviation. The typical composition of a hybrid-electric propulsion system comprises at least gas turbines, electric motors, energy storage, and an advanced power management system. In either case, the electric motor supplements the thrust of the gas turbine or operates in solo mode during select phases of flight, such as during taxiing and cruise conditions when power demands are low.

3.2 Benefits of Hybrid Electric Propulsion:

Optimization in the performance of the engines, which have to avoid high consumption of fossil fuel and offer quieter operations amongst others. Besides, it is growing with emerging technologies such as SOFC that add to systems and efficiently clean conversions to energy from hydrogen or other fuels. High-power-density light sources in aviation improved advanced thermal management systems and enhanced robust safety are the very important challenges for the introduction of hybrid-electric propulsion. In any case, continuous developments around materials, storage of energy, and system integration still keep some prospects open for regional and urban air mobility hybrid-electric applications.

3.3 Power requirements and flight phases:

- The highest power demands within the propulsion system occur during the take-off and climb phase; power may be directly available or augmented from a complementary system.
- At cruise, the power requirements are so much lower, and the system should be efficient at low output.
- There is a demand for power only in the decent; the system may, however, be used to add extra drag or recover energy.
- The system should be able to handle unforeseen difficulties, such as carrying reserves to make it safely to an alternate airport if necessary.
- Energy balance equations used for mission simulation to determine hydrogen mass and tank size as well as sizing of battery modules based on energy and power requirement.

3.4 Technical specifications for individual components used in the hybrid electric propulsion systems:

The hybrid electric propulsion system is made up of multiple components, with their potential technical specifications derived from research and study on such systems. These specifications can be applied to the proposed integration concepts as detailed below:

3.4.1 Turbofan Engine:

Gravimetric Power Density: 5.8 kW/kg

Mass: Between 5936–6120 kg for a Rolls-Royce Trent 1000

Volume: Length: 4.738 m, diameter: 2.85 m (fan)

3.4.2 SOFC:

Gravimetric Power Density: 8 kW/kg

Mass: 625 kg   125 kg

Volume: <0.5 m³ (estimated)

Electrical Efficiency: 50–60% (improved)

3.4.3 Thermal Management Systems:

Gravimetric Power Density: heat pipe prototypes transfer 700 W over compact geometries.

Volume: Compact designs with heat pipes as small as 4 mm × 120 mm cross-sections.

3.4.4 Hydrogen Tanks:

Gravimetric Energy Density: Liquid hydrogen energy density of 10.1 MJ/L, nearly 30% of kerosene's energy density.

Mass: Optimized tank (including hydrogen) ranges from 2719–4795 kg.

3.4.5 Electric Motor:

Gravimetric Power Density: Current systems: 5.2 kW/kg; Future potential: 10 kW/kg.

Mass: For a 27.6 MW system, motor mass is approximately 5269 kg

3.5 Why SOFC?

In general, a fuel cell (FC) is an electrochemical cell that uses the chemical potential of fuel to generate electrical energy through two redox reactions: an oxidation reaction and a reduction reaction. The FC directly transforms the chemical energy of fuel into electrical energy, in contrast to the combustion engine, which first transforms it into thermal energy, then mechanical energy, and finally electrical energy. Fuel cells come in a variety of forms, including solid oxide fuel cells (SOFC), direct methanol fuel cells (DMFC), alkaline fuel cells (AFC), Molten Carbonate Fuel Cell (MCFC) and polymer electrolyte membrane (PEM) fuel cells. Because of its potential benefits, which are discussed later in the study, SOFC has been taken into consideration in the suggested integration approaches.

The electrolyte of SOFC is yttrium-stabilized zirconia, while the anode is powered by hydrogen. Similar to the MCFC, an upstream reformer or, more often, an internal reformer may be used to extract H2 from hydrocarbons, mainly methane. The negatively charged oxygen ion O2 is transferred across the solid electrolyte. At the anode, it combines with the fuel to produce H2O and releases two electrons per H2 molecule. The SOFC's anode side is where the product water vapor must be eliminated.

SOFC operates at the highest temperature of any fuel cell type, they are a perfect candidate for internal reforming and Combined Heat and Power (CHP) applications. They have enormous potential for use in electrified propulsion powered by hydrogen. Additionally, they could allow for extremely effective long-distance electrified flight using SAF when combined with a gas engine. Nevertheless, SOFC have a very long startup time, and their high temperatures cause significant thermal corrosion and poor dynamic behaviour.

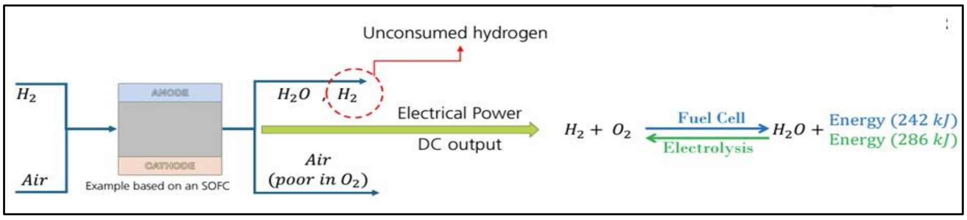

Fig1: SOFC Fuel Cell-HYDROGEN SYSTEM AND ARCHITECTURE INTEGRATION

Advantages of SOFC

- High efficiency
- High tolerance to contamination
- Fuel flexibility
- High operating temperatures
- Easier heat rejection due to high temperature gradient
- Long service life
- Technology readily available

Disadvantages of SOFC

- Medium Power Density
- Low start-stop and cycle stability
- Long startup time
- High degradation between electrolyte and electrode resulting in cracks

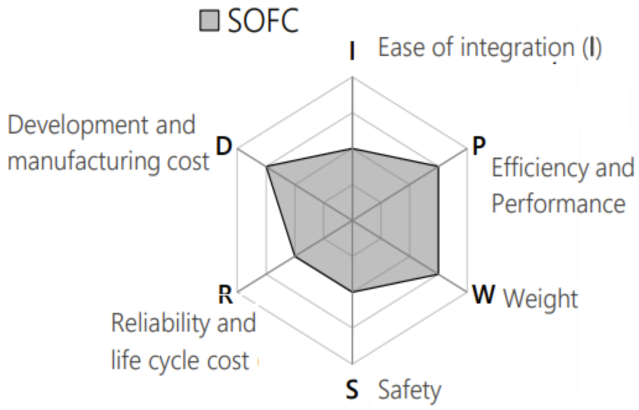

Fig 2: SOFC evaluation result

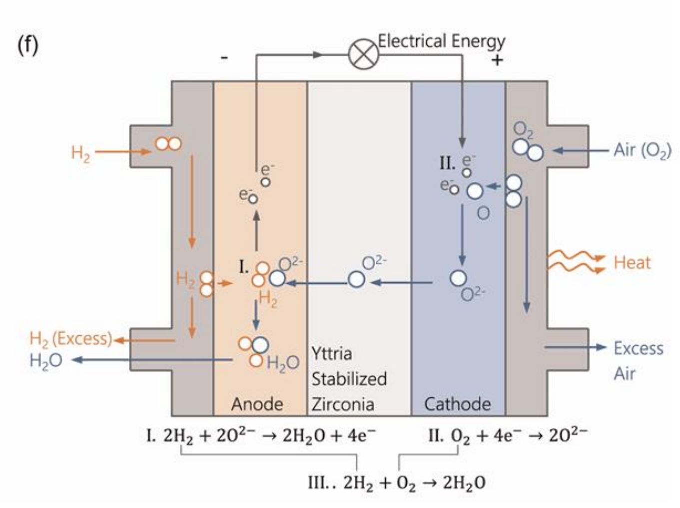

Fig 3: SOFC working principle

4. Proposed Integration concepts:

4.1. Hybrid-SOFC Turbofan Engine:

This proposed integration concept is inspired by the Rolls-Royce Trent 1000 high-bypass 3-shaft turbofan engine, which propels the Boeing 787 Dreamliner. This integration concept is influenced by the design and architecture of the Gulfstream G650 business jet. Commonly classified as the "tail-mounted engine" architecture. This type of architecture has its own potential benefits as mentioned later in the report. This concept consists of two turbofan engines mounted on the rear fuselage. The engine uses compressed hydrogen for combustion and fuel in an SOFC.

The SOFC in this integration concept operates at elevated pressure, as the pressurized hydrogen is supplied to the SOFC to move H_2 further to the combustion chamber. Hydrogen is an Ideal fuel that is used to reduce carbon footprints. It is a transition from traditional fuel to hydrogen use in combustion. When Hydrogen combusts, it releases water vapor as its By-Product. Apart from several advantages of using Hydrogen as a fuel for combustion, it has high energy density by mass which is nearly three times higher than the kerosene. The abundance and availability of hydrogen also make the preferred choice of fuel due to its ease of production through electrolysis.

Fig.4 Gulfstream G650

The integration concept includes the use of the following components:

- Compressed Hydrogen Tanks at the rear end of the fuselage
- Thermal Management System (TMS) of Hydrogen Tanks at the rear end of the fuselage
- Power Distribution Management System (PDMS) just beside the wings, distributing the electrical power to the auxiliary components, cockpit, and power electronics.
- Battery stacks stored at the Wings of the Aircraft

This concept is a nacelle-integrated gas turbine that includes the following components within the nacelle:

- LP Compressor (Fan)
- 8 Stages of IP Compressor
- 6 stages of HP Compressor
- SOFC with TMS
- Combustion chamber
- 1 stage of HP Turbine
- 1 stage of IP Turbine
- 6 Stages of LP Turbine
- Exhaust Nozzle

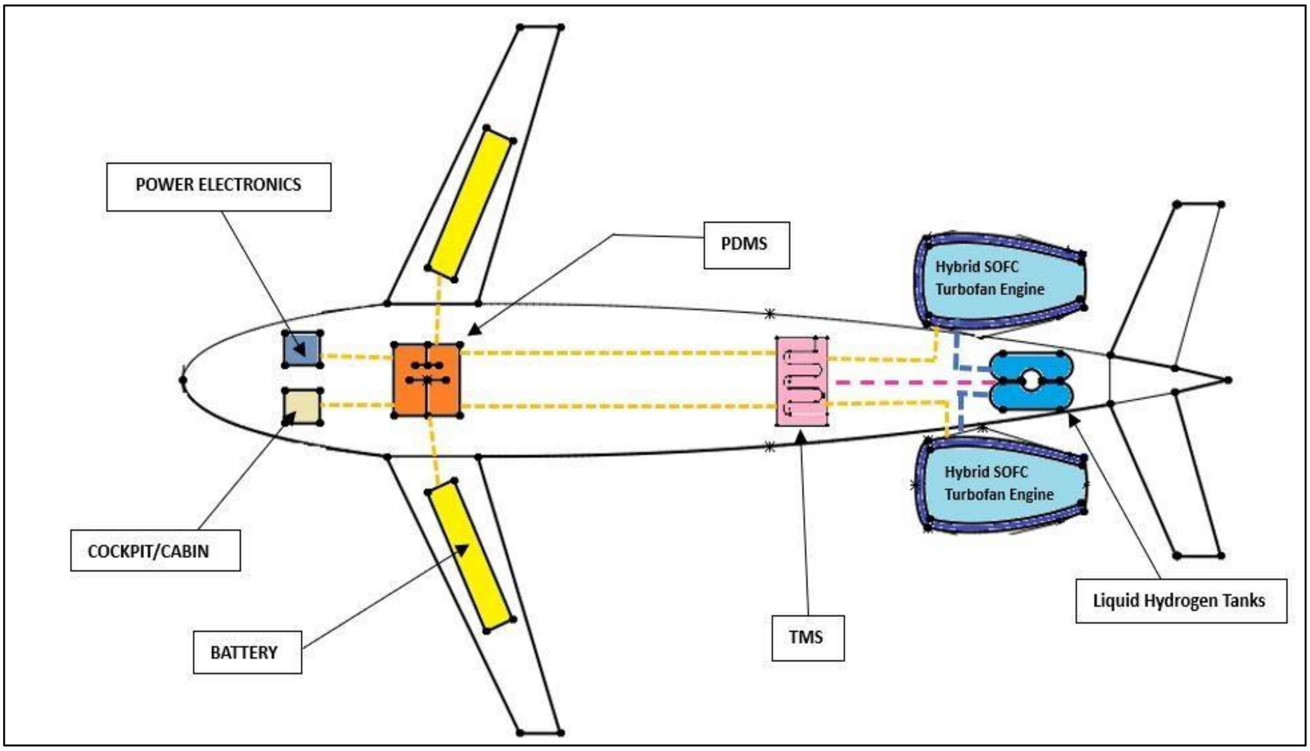

Fig 5. Top-view of the proposed Aeroengine Integration with the aircraft

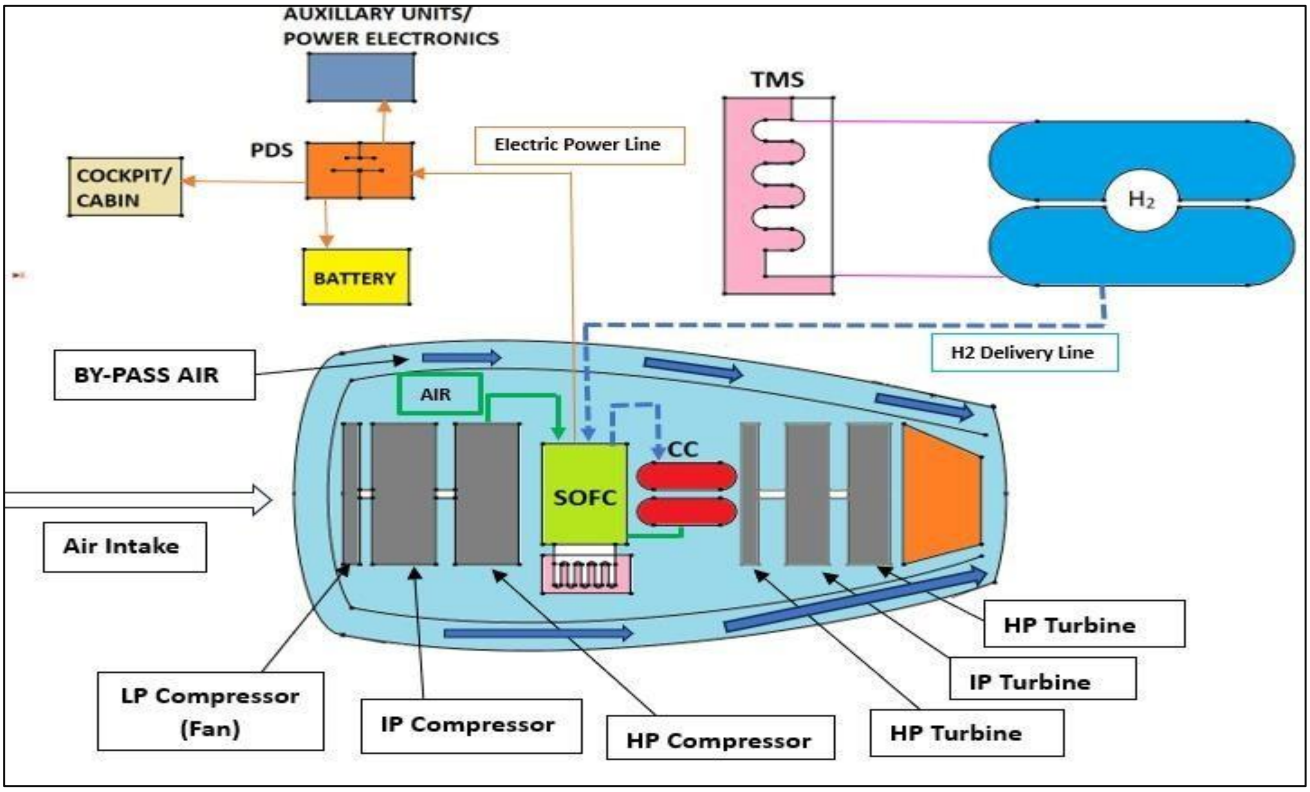

Fig 6. Engine Integration Hybrid SOFC Turbofan engine

Working:

- This nacelle-integrated gas turbine is a conventional gas turbine integrated with SOFC.
- The Air enters the engine core which gets further compressed by the several stages of compressors to SOFC.
- Compressed Hydrogen from tanks is supplied to SOFC at elevated pressures.
- SOFC operates at elevated pressures in this integration due to its potential benefits such as Thermodynamic Efficiency, Improved Heat Transfer properties, and Better Fuel Utilization.
- SOFC in turn produces Electricity which is further supplied to PDMS for the working of auxiliary components and power electronics.
- The remaining excess H_2 and Air is supplied to the Combustion chamber, where the chemical energy of the Fuel (H_2) is converted into Thermal Energy.
- This thermal energy is converted into Mechanical Energy through various stages of Turbines, hence producing thrust and propelling an aircraft.
- As similar to the high by-pass turbofan engine, the by-pass air plays a major role in the production of thrust.

Advantages of the proposed Integration Concept:

- The Aeroengine integration with the aircraft, which is two turbofan engines mounted at the rear end of the fuselage is inspired by the Gulfstream G650 and is classified as a 'tail-mounted engines' which has its own benefits such as Reduced cabin noise, increased safety in event of an engine fire, fan blade-off and improved aerodynamics.
- This proposed nacelle integrated gas turbine concept has a shorter hydrogen delivery line to the engine, due to which conditioning of hydrogen is easy as it is close to the engine at the rear. The fuel extraction strategy becomes easier.
- The Hydrogen leakage detection is a lot simpler due to the shorter delivery line.
- The primary and only fuel used in this engine for combustion and Fuel cells is Hydrogen. The additional weight of the Fuel such as Sustainable Aviation Fuel (SAF) or Kerosene has been eliminated.
- The battery stacks are placed in the wings, to maintain the stability of the Aircraft.
- No closer placement of electrical components, which reduces the Electromagnetic Interference (EMI).

Disadvantages of the proposed Integration Concept:

- Better thermal management is required for SOFC, because of its closeness to the combustion chamber.
- Larger nacelle volume to accommodate SOFC and TMS.
- Increased Structural Weight due to engines mounted on the rear side.
- Low volumetric density of hydrogen as a fuel for combustion as compared to SAF.
- Additional components are required to provide pressurized Hydrogen to SOFC, and to maintain the ambient pressure to the combustion chamber from SOFC.
- Infrastructure requirements for combustion of Hydrogen.
- Poor quality of Air (O2) exiting the fuel cell before combustion may affect the combustion process.
- SOFC-related challenges also play a vital role.

4.2. Partially Nacelle Integrated Propulsion System:

The placement of components in a hybrid electric system is complex and for achieving the optimal performance and reliability with safety considerations. In this design, the aircraft will consist of mainly a Gas turbine with a generator, fuel cell stack, electric components, thermal management system, Hydrogen storage tanks, power electronics, gearbox, and Sustainable Aviation Fuel (SAF). This design makes use of both SOFC and SAF for propulsion. The source of power is switched based on the power requirements and operating conditions. This integration concept has a two turbo-prop engine mounted on wings consisting gas turbine integrated with SOFC.

The propeller-driven gas turbine engine with electric motor which is integrated in a Nacelle is mounted on the wing for its aerodynamic efficiencies and optimising airflow. The SOFC is also a secondary source of power apart from gas turbines to produce power. Similar to the conventional aircraft, the SAF is stored in the wings and the hydrogen tanks are at the rear end of the fuselage (Tail). Locating the Hydrogen tanks in the tail creates a physical barrier between hydrogen storage and the cabin, reducing the damage and effect in case of any system failure. The tail can be enhanced with structural reinforcements and fire-resistant materials.

The integration concept includes the use of the following components:

- Compressed Hydrogen Tanks at the rear end of the fuselage (Tail)
- Thermal Management System (TMS) of Hydrogen Tanks at the rear end of the fuselage
- Power Distribution Management System (PDMS) just beside the wings, distributing the electrical power to the auxiliary components, cockpit, and power electronics.
- Battery stacks placed inside the nacelle mounted on the wings.
- Fuel Tanks (SAF) at the wings
- Propeller
- Gas Turbine with Generator (Compressor + Combustion Chamber + Turbine)
- Electrical Motor
- Battery
- SOFC with TMS
- Exhaust Nozzle

Working:

- The system features a single-shaft turboprop engine integrated with a SOFC within the nacelle.
- This aero-engine has two power-producing sources: a gas turbine with a generator and a SOFC.
- The primary concept behind this integration is that, during high-power demand conditions such as take-off and climb, power is drawn from the gas turbine using conventional SAF (Sustainable Aviation Fuel). In contrast, during cruise and other low-power-demand phases, power is supplied by the SOFC fuel cell.
- A battery serves as a redundancy for the system.
- The conventional gas turbine, integrated with the generator, draws in air, compresses it, and combusts it. This drives the turbine, generating electricity, which in turn powers the propeller through an electric motor.
- During take-off, power is drawn from the gas turbine, but once power demands decrease, the gas turbine is shut off, and the fuel cell begins supplying power during cruise and other phases requiring lower power.
- Compressed hydrogen from the tanks is supplied to the SOFC during the cruise phase of flight.
- The electricity produced by the SOFC is used to power the Power Distribution Management System (PDMS), which supports the operation of auxiliary components and power electronics.

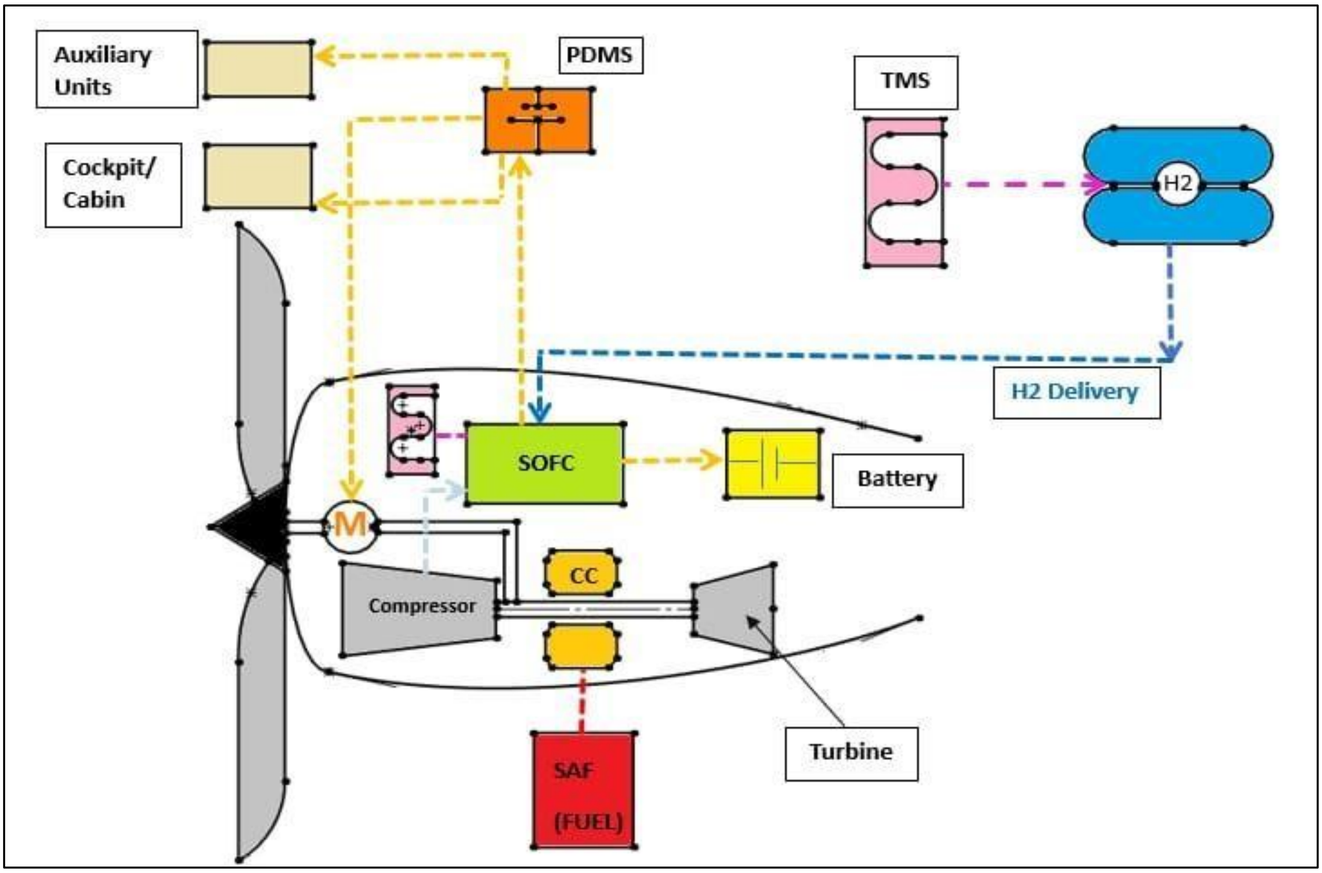

Fig 6. Engine Integration

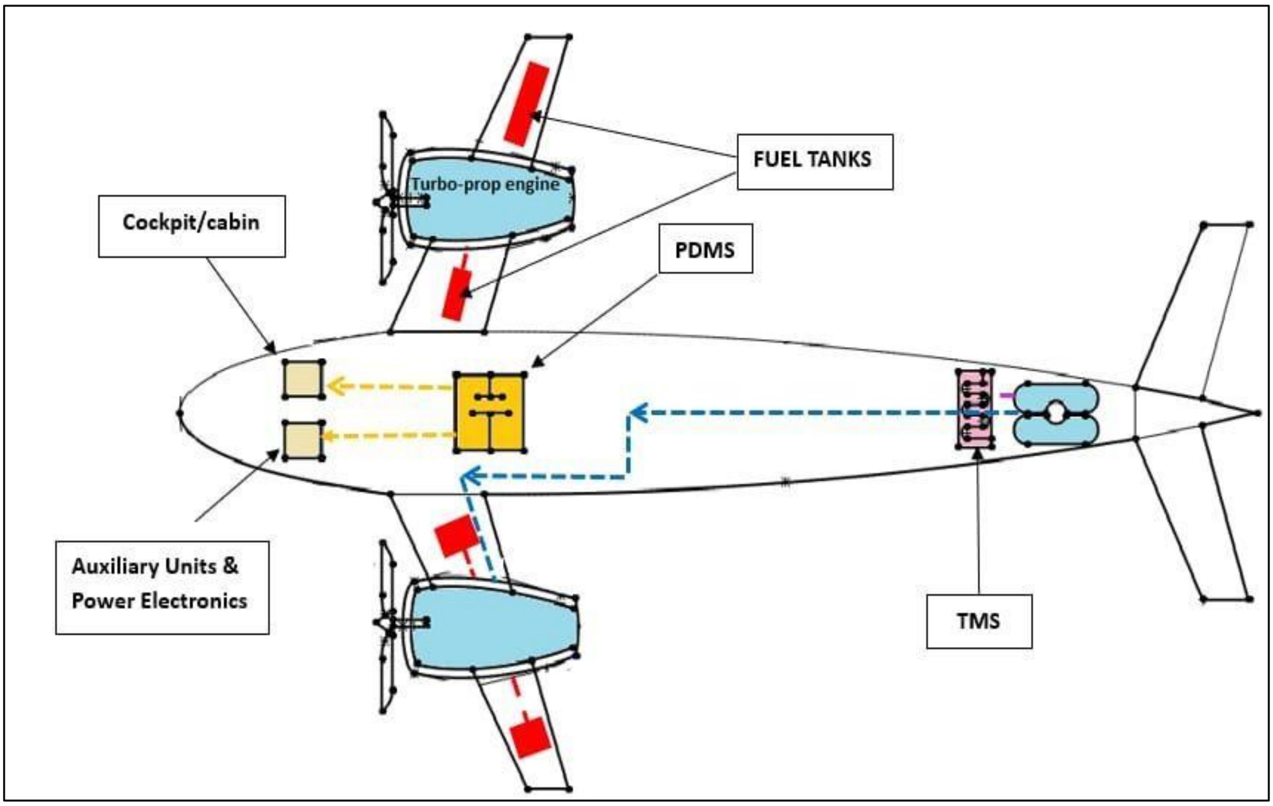

Fig 7. Top-view of the proposed Aeroengine Integration with the aircraft

Advantages of the proposed integration concept

- Placing a gas turbine and nacelle on wing improves aerodynamic flow enhances performance and reduces drag. This integration has a better distribution of weight and also provides balance for aircraft.
- Locating the hydrogen tanks at the tail creates a barrier between passengers and certain leakages or system failures. Besides, at the tail of the aircraft is space to house large cylindrical tanks without compromising on cabin or cargo space. A tail can also be designed and made with fire-resistant materials to handle the particular challenges of hydrogen storage.
- Keeping the storage of hydrogen off the fuselage simplifies cabin design, enhances passenger safety, and maximizes cargo capacity.
- The design allows for easier scaling to different aircraft sizes and applications, especially as hydrogen technology.
- Fewer powertrain components in fuselage which leads to improved zonal and functional separation between cabin and electrical components.
- Increased wing mass, hence resulting in less stress in the wing structure.
- A significant amount of fuel is conserved during the low power requirements and operating conditions of the aircraft.

Disadvantages of the proposed integration concept:

- Hydrogen tanks at the tail may cause a center-of-gravity shift, requiring careful design to have a stable flight. Weight at the tail may require a larger horizontal stabilizer, increasing overall drag.
- SOFC stacking in the nacelle can result in complicated nacelle designs.
- Routing hydrogen from the tail to the SOFC and gas turbine in the wing requires long, insulated pipelines that increase system weight and leakage points. Additional thermal insulation and pressurization may be needed to reduce the hydrogen losses.
- Weight added to reinforce the tail for heavy hydrogen tanks, fire-resistant materials, and insulation further raise the structural demands and costs.
- Tail-mounted hydrogen tanks are generally less accessible to maintenance, inspection, and refueling compared with tanks installed in fuselage locations; Wing-integrated SOFC stacks may complicate the wing maintenance concept.
- Wing-mounted nacelles, though being aerodynamically efficient, can introduce drag if the design is not considering necessary conditions and situations. The increased wing loading from the SOFC stack may have an impact on the flexibility and performance of the wing.
- Hydrogen tanks at the tail may be much more vulnerable during a crash even with suitable reinforcements.

5. Conclusion:

The integration of SOFC with a turbofan engine (Concept 1) and the concept of partial nacelle integration (Concept 2) both offer distinct benefits and drawbacks in terms of their performance, safety, and feasibility. These pros and cons intensively depend on the mission profile and aircraft requirements. In addition, the SOFC-turbofan integration will be more adequate for medium to long-range missions, while the partial nacelle concept will be more suitable for small aircraft, as those designed for regional or urban air mobility. Whereas turboprop engines are normally used in small jets or chartered aircraft, turbofans are mainly applied to large civil aircraft, like wide-body ones.

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